

Profile

Senior software engineer with 18 years of experience building production systems across backend services and robotics software.

Strong background in Python and C++, with a focus on system design, reliability, and long-term maintainability. Experienced in taking complex systems from research or prototype stages into stable, real-world deployment.

Skills

Core Languages

C++, Python

Backend & Distributed Systems

FastAPI, REST, Kafka, RabbitMQ, Temporal, MongoDB

Robotics Software

ROS, ROS2, SLAM, Gazebo, PX4

Infrastructure & Tooling

Docker, Linux, Git

Operating Systems

Linux (primary), Windows

Education

Ph.D.

Electrical engineering
Technical University of Košice
📅 09/2020 – 08/2023

M.S.

Computer Science
Technical University of Košice
📅 09/2003 – 06/2008

Secondary Technical School of
Electrical Engineering

📅 09/1999 – 06/2003

Experience

AT&T

📅 01/2024 – PRESENT

Senior Full Stack Development Engineer

- Developed and maintained a distributed order management system handling the full customer order lifecycle, from creation to fulfillment.
- Implemented backend services for processing Kafka events, coordinating workflows with Temporal, and persisting state in MongoDB.

Python, FastAPI, Kafka, Temporal, MongoDB

Photoneo / Brightpick

📅 10/2019 – 01/2024

Senior software developer

- Designed and implemented fleet management software for automated warehouses, enabling coordination and control of multiple autonomous mobile robots.
- Developed core robot software, including APIs, sensor integration, hardware communication, and localization using ROS.
- Built an autonomous indoor 3D-mapping drone for GPS-denied environments as part of PhD research, covering navigation, mapping, and system integration.

C++, Python, ROS, ROS2, Docker, SLAM, PX4, RTAB-Map

ComAp

📅 10/2018 – 10/2019

Embedded programmer

- Developed embedded control software for power generation systems, including a PLC interpreter and a MODBUS client used for device communication.

C, MODBUS

ESET

📅 09/2008 – 10/2018

Software developer

- Developed a push notification system, including client-side components and an MQTT protocol implementation.
- Maintained integration of an antispam engine within the ESET Smart Security product.
- Designed and implemented a DNS tunneling module for data transfer in restricted network environments.
- Built a web-based system for collecting, processing, and visualizing antispam statistics.

C++, C, Python, MQTT, SQL, Qt